Top 100 Multiple Choice Questions with Answers on Basics of Big Data

1. Question: What does the term "big data" refer to?
a) Any large dataset
b) Data that cannot be processed
c) Data that is too complex to analyze
d) Large and complex datasets that require specialized tools and techniques
Answer: d) Large and complex datasets that require specialized tools and techniques

2. Question: What are the three main characteristics of big data known as the "Three Vs"?
a) Volume, Variety, Velocity
b) Volume, Value, Vulnerability
c) Veracity, Velocity, Variety
d) Value, Variety, Velocity
Answer: a) Volume, Variety, Velocity

3. Question: Which term refers to the process of analyzing large datasets to uncover hidden patterns and insights?
a) Data warehousing
b) Data mining
c) Data storage
d) Data aggregation
Answer: b) Data mining

4. Question: What is the primary goal of data preprocessing in big data analysis?
a) To increase the size of the dataset
b) To reduce the volume of the dataset
c) To enhance the quality of the dataset
d) To eliminate variety in the dataset
Answer: c) To enhance the quality of the dataset

5. Question: What is the role of Hadoop in big data processing?
a) Hadoop is a programming language for big data analysis
b) Hadoop is a type of database used for big data storage
c) Hadoop is a framework for distributed processing of large datasets
d) Hadoop is a visualization tool for big data analysis
Answer: c) Hadoop is a framework for distributed processing of large datasets

mcqxpert.blogspot.com

6. Question: Which programming language is commonly used for big data analysis and processing?
a) Java
b) Python
c) C++
d) Ruby
Answer: b) Python

7. Question: What is the purpose of MapReduce in Hadoop?
a) To create maps of geographical locations
b) To visualize data on maps
c) To process and analyze large datasets in parallel
d) To generate reports from data
Answer: c) To process and analyze large datasets in parallel

8. Question: What is the main advantage of using distributed storage systems in big data environments?
a) Centralized management of data
b) Faster data processing speed
c) Lower cost of storage
d) Redundancy and fault tolerance
Answer: d) Redundancy and fault tolerance

9. Question: Which type of data refers to information that is generated in real-time and requires immediate processing?
a) Structured data
b) Semi-structured data
c) Unstructured data
d) Streaming data
Answer: d) Streaming data

10. Question: What is the purpose of data partitioning in big data processing?
a) To remove irrelevant data
b) To distribute data across multiple storage devices
c) To merge data from different sources
d) To visualize data patterns
Answer: b) To distribute data across multiple storage devices

11. Question: What is the term for the process of extracting valuable insights and information from raw data?
a) Data storage
b) Data mining
c) Data aggregation

mcqxpert.blogspot.com

d) Data cataloging
Answer: b) Data mining

12. Question: Which term describes data that is generated by machines, sensors, or devices?
a) Human-generated data
b) User-generated data
c) Machine-generated data
d) Process-generated data
Answer: c) Machine-generated data

13. Question: What is the purpose of a data lake in big data architecture?
a) To store only structured data
b) To store data in a structured format
c) To store data in a single database
d) To store raw and unstructured data for future analysis
Answer: d) To store raw and unstructured data for future analysis

14. Question: Which big data processing framework is designed for real-time stream processing?
a) Hadoop
b) Apache Spark
c) Apache Kafka
d) MongoDB
Answer: c) Apache Kafka

15. Question: What is the concept of "Data Governance" in big data?
a) Maximizing data storage capacity
b) Ensuring data quality, security, and compliance
c) Sharing data with external partners
d) Reducing data variety
Answer: b) Ensuring data quality, security, and compliance

16. Question: Which type of database is optimized for storing and querying graph-like data structures?
a) Relational database
b) Document database
c) Graph database
d) Key-value database
Answer: c) Graph database

17. Question: What does the term "data silo" refer to in the context of big data?

mcqmert.blogspot.com

a) A storage unit for structured data
b) A centralized repository for all data types
c) Isolated and disconnected data storage systems
d) A type of data visualization technique
Answer: c) Isolated and disconnected data storage systems

18. Question: Which big data technology uses parallel processing for distributed data storage and computation?
a) Hadoop
b) SQL Server
c) Tableau
d) MongoDB
Answer: a) Hadoop

19. Question: What is the main advantage of using in-memory databases for big data processing?
a) Lower cost of storage
b) Slower data processing speed
c) Enhanced data durability
d) Faster data retrieval and analysis
Answer: d) Faster data retrieval and analysis

20. Question: What is the purpose of data deduplication in big data storage?
a) To increase data variety
b) To reduce data volume by eliminating duplicate records
c) To create data backups
d) To convert unstructured data into structured format
Answer: b) To reduce data volume by eliminating duplicate records

21. Question: Which data visualization tool is widely used for creating interactive and dynamic visualizations of big data?
a) Microsoft Excel
b) Tableau
c) Power BI
d) Google Sheets
Answer: b) Tableau

22. Question: What is "ETL" in the context of big data processing?
a) Extract, Transform, Load – a process to ingest, process, and analyze data
b) Efficient Time Logging – a technique for tracking data usage
c) Early Termination Logic – a method to stop data processing early
d) Extended Transformation Layer – a data storage architecture
Answer: a) Extract, Transform, Load – a process to ingest, process, and analyze data

mcqxpert.blogspot.com

23. Question: What is the main purpose of a data warehouse in the big data ecosystem?
a) To store raw and unprocessed data
b) To store data in its original format
c) To aggregate and store data from various sources for analysis
d) To visualize data in real-time
Answer: c) To aggregate and store data from various sources for analysis

24. Question: Which data storage technology offers a distributed file system that provides high availability and fault tolerance?
a) Hadoop Distributed File System (HDFS)
b) Network Attached Storage (NAS)
c) Solid State Drive (SSD)
d) Hierarchical Storage Management (HSM)
Answer: a) Hadoop Distributed File System (HDFS)

25. Question: What is the purpose of data anonymization in big data analysis?
a) To increase data volume for analysis
b) To preserve individual privacy by removing or altering personal information
c) To aggregate data into summarized format
d) To convert unstructured data into structured format
Answer: b) To preserve individual privacy by removing or altering personal information

26. Question: Which data processing technique allows for more flexible and interactive exploration of data?
a) Batch processing
b) Stream processing
c) Interactive processing
d) Sequential processing
Answer: c) Interactive processing

27. Question: What is "data lineage" in the context of big data governance?
a) A technique for storing data in a single lineage
b) The process of organizing data silos
c) The history and tracking of data movement and transformations
d) The process of transforming structured data into unstructured format
Answer: c) The history and tracking of data movement and transformations

28. Question: Which type of data analysis focuses on exploring data to discover new patterns and insights?
a) Descriptive analysis
b) Diagnostic analysis
c) Predictive analysis

mcqxpert.blogspot.com

d) Exploratory analysis
Answer: d) Exploratory analysis

29. Question: What is the main advantage of using a columnar database for big data storage and analysis?
a) Faster data insertion speed
b) Better data compression and query performance
c) Enhanced data durability
d) Lower cost of storage
Answer: b) Better data compression and query performance

30. Question: What does the term "Data Lakehouse" refer to in the context of big data architecture?
a) A storage solution exclusively for structured data
b) A unified approach that combines data lakes and data warehouses
c) A specialized database for machine-generated data
d) A virtual repository for streaming data
Answer: b) A unified approach that combines data lakes and data warehouses

31. Question: Which big data concept involves the use of external third-party data to enhance insights?
a) Data augmentation
b) Data bypass
c) Data sideloading
d) Data offloading
Answer: a) Data augmentation

32. Question: What is the primary objective of a data steward in the context of big data governance?
a) Data visualization
b) Data deletion
c) Data transformation
d) Data quality and compliance
Answer: d) Data quality and compliance

33. Question: Which type of machine learning algorithm is suitable for solving classification problems with big data?
a) Decision trees
b) Linear regression
c) K-means clustering
d) Association rules
Answer: a) Decision trees

mcqxpert.blogspot.com

34. Question: What is the significance of the "Lambda Architecture" in big data processing?
a) A programming language for big data analysis
b) A design pattern that combines batch and stream processing
c) A specialized database for graph data
d) A technique for real-time data visualization
Answer: b) A design pattern that combines batch and stream processing

35. Question: What is the purpose of a "Data Mart" in a big data environment?
a) To store raw and unprocessed data
b) To store data temporarily for analysis
c) To provide specialized data for specific user groups
d) To visualize data in real-time
Answer: c) To provide specialized data for specific user groups

36. Question: Which big data technology enables real-time analysis of complex event streams?
a) Apache HBase
b) Apache Cassandra
c) Apache Flink
d) Apache Pig
Answer: c) Apache Flink

37. Question: In big data terminology, what is "Feature Engineering"?
a) A process to generate new features from existing data
b) A technique to compress large datasets
c) A method for visualizing data patterns
d) A process to remove irrelevant data
Answer: a) A process to generate new features from existing data

38. Question: Which cloud service model offers the highest level of control and customization for big data processing?
a) Infrastructure as a Service (IaaS)
b) Platform as a Service (PaaS)
c) Software as a Service (SaaS)
d) Function as a Service (FaaS)
Answer: a) Infrastructure as a Service (IaaS)

39. Question: What is the main purpose of "Dimensionality Reduction" in big data analysis?
a) To increase data variety
b) To expand the dataset size

mcqxpert.blogspot.com

c) To reduce the number of features while preserving relevant information
d) To create high-dimensional visualizations
Answer: c) To reduce the number of features while preserving relevant information

40. Question: Which big data concept involves the use of data from social media platforms, forums, and blogs for analysis?
a) Social data analytics
b) Sentiment analysis
c) Social data harvesting
d) Social media integration
Answer: a) Social data analytics

41. Question: What is the term for a data quality issue in which different data sources use different units of measurement?
a) Data inconsistency
b) Data duplication
c) Data integration
d) Data conversion
Answer: d) Data conversion

42. Question: Which data storage technology is designed for high-speed data ingestion and real-time analytics?
a) Hadoop HDFS
b) Apache Kafka
c) Amazon S3
d) Google Cloud Storage
Answer: b) Apache Kafka

43. Question: In big data processing, what does "CAP Theorem" refer to?
a) A theorem related to data encryption
b) A theorem related to data visualization
c) A theorem related to data compression
d) A theorem related to data consistency, availability, and partition tolerance
Answer: d) A theorem related to data consistency, availability, and partition tolerance

44. Question: Which big data storage technology is specifically designed for handling large volumes of time-series data?
a) Cassandra
b) Hadoop HDFS
c) InfluxDB
d) MongoDB
Answer: c) InfluxDB

mcqmcq.blogspot.com

45. Question: What is "Natural Language Processing" (NLP) in the context of big data?
a) A process for extracting features from text data
b) A technique for compressing large text datasets
c) A method for visualizing text patterns
d) A process for transforming unstructured text into structured format
Answer: a) A process for extracting features from text data

46. Question: Which type of data storage technology is optimized for storing and querying geospatial data?
a) NoSQL databases
b) Columnar databases
c) Graph databases
d) Document databases
Answer: c) Graph databases

47. Question: What is "Data Lineage" in the context of big data governance?
a) A method for tracing the history and transformation of data
b) A technique for encrypting data at rest
c) A process for data deduplication
d) A technology for real-time data synchronization
Answer: a) A method for tracing the history and transformation of data

48. Question: Which data analysis technique involves finding associations and relationships between variables in a dataset?
a) Clustering analysis
b) Regression analysis
c) Association rule mining
d) Anomaly detection
Answer: c) Association rule mining

49. Question: What is "Data Curation" in the context of big data management?
a) A process for data warehousing
b) A process for data transformation
c) A process for data cleaning, enrichment, and maintenance
d) A process for data visualization
Answer: c) A process for data cleaning, enrichment, and maintenance

50. Question: What is the concept of "Dark Data" in the realm of big data?
a) Data that is intentionally hidden from analysis
b) Data that is difficult to read and interpret
c) Data that is unused and remains untapped for insights
d) Data that is encrypted and inaccessible

mcqxpert.blogspot.com

Answer: c) Data that is unused and remains untapped for insights

---

51. Question: What is the primary goal of "Data Ingestion" in big data processing?
a) To exclude irrelevant data
b) To transform unstructured data into structured format
c) To load and prepare data for analysis
d) To create visualizations from raw data
Answer: c) To load and prepare data for analysis

52. Question: Which technology is commonly used for real-time analysis of log data in big data applications?
a) Hadoop
b) Spark
c) ELK Stack (Elasticsearch, Logstash, Kibana)
d) MongoDB
Answer: c) ELK Stack (Elasticsearch, Logstash, Kibana)

53. Question: What is the purpose of "Data Reservoir" in big data architecture?
a) A storage area for high-value data
b) A storage area for frequently accessed data
c) A centralized repository for raw and unstructured data
d) A repository for summarized and aggregated data
Answer: c) A centralized repository for raw and unstructured data

54. Question: Which big data technology is designed for handling and analyzing time-series data from IoT devices?
a) Apache Spark
b) Apache Cassandra
c) Apache Kafka
d) Apache HBase
Answer: c) Apache Kafka

55. Question: What is the concept of "Data Sovereignty" in the context of big data governance?
a) The right to own and control one's personal data
b) The global sharing of data without restrictions
c) The responsibility to delete all data after a certain period
d) The data's ability to self-govern its storage and processing
Answer: a) The right to own and control one's personal data

56. Question: Which data processing framework focuses on providing low-latency interactive query capabilities for big data?

mcqxpert.blogspot.com

a) Hadoop
b) Apache Flink
c) Apache Hive
d) Apache Pig
Answer: c) Apache Hive

57. Question: In big data analytics, what is the term for the process of combining data from different sources into a single dataset?
a) Data transformation
b) Data integration
c) Data augmentation
d) Data aggregation
Answer: b) Data integration

58. Question: Which machine learning technique involves grouping similar data points together based on their characteristics?
a) Regression
b) Clustering
c) Classification
d) Anomaly detection
Answer: b) Clustering

59. Question: What is the primary purpose of "Data Masking" in big data security?
a) To hide data from unauthorized users
b) To convert unstructured data into structured format
c) To reduce data volume for analysis
d) To visualize data patterns
Answer: a) To hide data from unauthorized users

60. Question: Which data storage technology is optimized for storing and querying graph data structures?
a) Hadoop HDFS
b) Apache Cassandra
c) Neo4j
d) Amazon S3
Answer: c) Neo4j

61. Question: What is the term for a data quality issue in which data values are missing or incomplete?
a) Data inconsistency
b) Data redundancy
c) Data integrity
d) Data imputation

mcqxpert.blogspot.com

Answer: d) Data imputation

62. Question: Which big data concept focuses on the practice of treating data as a corporate asset and assigning ownership?
a) Data governance
b) Data democracy
c) Data anarchy
d) Data socialism
Answer: a) Data governance

63. Question: What is the purpose of "Schema-on-Read" in contrast to "Schema-on-Write" in big data storage?
a) Schema-on-Read is used for data storage, while Schema-on-Write is used for data analysis
b) Schema-on-Read defines the data structure during analysis, while Schema-on-Write defines it during storage
c) Schema-on-Read simplifies data storage, while Schema-on-Write simplifies data analysis
d) Schema-on-Read and Schema-on-Write are two terms for the same data processing approach
Answer: b) Schema-on-Read defines the data structure during analysis, while Schema-on-Write defines it during storage

64. Question: What is "Federated Query" in the context of big data processing?
a) A technique to query multiple data sources as if they were a single database
b) A query language for analyzing structured data
c) A query optimization technique for stream processing
d) A method to divide queries into smaller federated segments
Answer: a) A technique to query multiple data sources as if they were a single database

65. Question: Which data processing approach combines batch processing and stream processing for real-time insights?
a) Lambda Architecture
b) Sigma Architecture
c) Theta Architecture
d) Kappa Architecture
Answer: a) Lambda Architecture

66. Question: What is "Polyglot Persistence" in the context of big data storage?
a) The practice of using a single database for all types of data
b) The use of multiple programming languages for data processing
c) The use of different databases optimized for different types of data
d) The practice of storing data in plain text format for simplicity
Answer: c) The use of different databases optimized for different types of data

mcqpet.blogspot.com

67. Question: Which data processing approach involves creating intermediate summaries of data to speed up query performance?
a) Online Analytical Processing (OLAP)
b) Online Transaction Processing (OLTP)
c) Data Warehousing
d) Data Mining
Answer: a) Online Analytical Processing (OLAP)

68. Question: What is the primary advantage of using "Parquet" or "ORC" file formats for big data storage?
a) Smaller file sizes and improved query performance
b) Support for real-time streaming data
c) Enhanced data durability and replication
d) Compatibility with relational databases
Answer: a) Smaller file sizes and improved query performance

69. Question: Which big data technology is designed for processing and analyzing large amounts of text data?
a) Natural Language Processing (NLP)
b) Text-to-Speech (TTS) conversion
c) Optical Character Recognition (OCR)
d) Machine Learning
Answer: a) Natural Language Processing (NLP)

70. Question: In the context of big data, what does "Data Gravity" refer to?
a) The tendency of data to be attracted to large storage solutions
b) The concentration of data in certain geographic regions due to latency concerns
c) The mass or size of a dataset
d) The force that pulls data towards analytical tools
Answer: b) The concentration of data in certain geographic regions due to latency concerns

71. Question: What is the concept of "Data Provenance" in big data analysis?
a) A technique for validating the authenticity of data
b) The process of transforming raw data into usable insights
c) The history and origin of data, including its movement and transformations
d) A method for visualizing the flow of data within a network
Answer: c) The history and origin of data, including its movement and transformations

72. Question: Which big data technology provides an interactive and visual environment for data exploration and analysis?
a) Apache Flink
b) Apache Spark

mcqxpert.blogspot.com

c) Tableau
d) Apache HBase
Answer: c) Tableau

73. Question: What is the primary purpose of "Schema Evolution" in big data processing?
a) To create a unified schema for all data sources
b) To modify the data schema without disrupting existing processes
c) To standardize data formats across different databases
d) To merge multiple datasets into a single schema
Answer: b) To modify the data schema without disrupting existing processes

74. Question: Which concept involves the analysis of data generated by Internet of Things (IoT) devices?
a) Edge computing
b) Fog computing
c) Mist computing
d) Rain computing
Answer: a) Edge computing

75. Question: What is "Data Virtualization" in the context of big data architecture?
a) The process of transforming raw data into structured format
b) A technique for creating virtual replicas of physical data sources
c) The practice of visualizing data using virtual reality technology
d) The process of distributing data across virtual machines
Answer: b) A technique for creating virtual replicas of physical data sources

76. Question: Which data processing technique focuses on identifying patterns and trends in data over time?
a) Time series analysis
b) Cross-sectional analysis
c) Longitudinal analysis
d) Exploratory analysis
Answer: a) Time series analysis

77. Question: What is "Data Wrangling" in the context of big data preparation?
a) A technique for data encryption
b) The process of creating data visualizations
c) The process of cleaning, transforming, and mapping data for analysis
d) A method for data compression
Answer: c) The process of cleaning, transforming, and mapping data for analysis

mcqpert.blogspot.com

78. Question: Which big data technology enables the processing of complex SQL queries on large datasets?
a) Apache Cassandra
b) Apache Hadoop
c) Google BigQuery
d) Apache Kafka
Answer: c) Google BigQuery

79. Question: What is the concept of "Data Exfiltration" in the context of big data security?
a) The process of extracting valuable insights from raw data
b) The unauthorized removal of data from a network or system
c) The practice of encrypting data at rest
d) The process of transforming unstructured data into structured format
Answer: b) The unauthorized removal of data from a network or system

80. Question: Which technology is used to store and manage data in a distributed and fault-tolerant manner across multiple nodes?
a) Distributed Ledger Technology (DLT)
b) Blockchain
c) Consensus algorithm
d) Raft algorithm
Answer: a) Distributed Ledger Technology (DLT)

81. Question: What is "Data Lineage" in the context of big data governance?
a) A technique for tracking data movement between different storage locations
b) The history and tracking of data changes and transformations
c) The process of merging different datasets into a single source
d) The process of loading data into a data warehouse
Answer: b) The history and tracking of data changes and transformations

82. Question: Which machine learning technique focuses on making predictions based on historical data patterns?
a) Clustering
b) Regression
c) Classification
d) Anomaly detection
Answer: b) Regression

83. Question: What is "Data Catalog" in the context of big data management?
a) A physical repository for storing data
b) A tool for data visualization
c) A centralized metadata repository for data assets

mcqxpert.blogspot.com

d) A technique for data encryption
Answer: c) A centralized metadata repository for data assets

84. Question: Which data storage technology is optimized for storing and querying time-series data related to business metrics?
a) Apache Cassandra
b) InfluxDB
c) Amazon Redshift
d) Apache HBase
Answer: b) InfluxDB

85. Question: What is the primary goal of "Data Fusion" in big data processing?
a) To separate data into distinct categories
b) To combine data from multiple sources to generate more accurate insights
c) To compress data for storage efficiency
d) To visualize data patterns using fusion techniques
Answer: b) To combine data from multiple sources to generate more accurate insights

86. Question: In the context of big data analytics, what is "Interpolation"?
a) The process of reducing data volume for analysis
b) The practice of predicting missing values based on existing data points
c) The process of extracting data from external sources
d) The practice of creating visualizations from raw data
Answer: b) The practice of predicting missing values based on existing data points

87. Question: Which big data technology is designed for processing and analyzing data streams from social media platforms?
a) Apache Flink
b) Apache Kafka
c) Apache HBase
d) Apache Hive
Answer: b) Apache Kafka

88. Question: What is the concept of "Data Resilience" in the context of big data?
a) The ability of data to recover from hardware failures
b) The practice of replicating data across multiple locations for disaster recovery
c) The process of converting unstructured data into structured format
d) The practice of data compression for storage efficiency
Answer: b) The practice of replicating data across multiple locations for disaster recovery

89. Question: Which data processing technique involves applying statistical algorithms to identify relationships between variables?
a) Exploratory analysis

mcqxpert.blogspot.com

b) Regression analysis
c) Time series analysis
d) Clustering analysis
Answer: b) Regression analysis

90. Question: What is the concept of "Data Gravity" in the context of big data?
a) The tendency of data to be attracted to large storage solutions
b) The concentration of data in certain geographic regions due to latency concerns
c) The mass or size of a dataset
d) The force that pulls data towards analytical tools
Answer: b) The concentration of data in certain geographic regions due to latency concerns

91. Question: What is the primary purpose of "Data Curation" in big data management?
a) To store data in its raw and unprocessed form
b) To ensure data quality, accuracy, and consistency
c) To encrypt data for security purposes
d) To visualize data patterns using charts and graphs
Answer: b) To ensure data quality, accuracy, and consistency

92. Question: Which data processing technique involves analyzing data based on geographical and spatial relationships?
a) Temporal analysis
b) Spatial analysis
c) Frequency analysis
d) Causal analysis
Answer: b) Spatial analysis

93. Question: What is the primary goal of "Data Imputation" in big data analysis?
a) To create a unified schema for all data sources
b) To merge multiple datasets into a single source
c) To predict missing values in a dataset based on available information
d) To reduce data redundancy and duplication
Answer: c) To predict missing values in a dataset based on available information

94. Question: Which concept focuses on the practice of analyzing data as it is generated, in real-time?
a) Batch processing
b) Stream processing
c) Incremental processing
d) Sequential processing
Answer: b) Stream processing

mcqnert.blogspot.com

95. Question: In the context of big data, what does "Data Anonymization" refer to?
a) The process of compressing large datasets for efficient storage
b) The practice of transforming unstructured data into structured format
c) The process of removing or altering personal information to protect privacy
d) The technique of visualizing data patterns using graphical representations
Answer: c) The process of removing or altering personal information to protect privacy

96. Question: Which big data technology is commonly used for interactive querying and analysis of large datasets?
a) Apache Spark
b) Apache Kafka
c) Apache Flink
d) Apache Pig
Answer: a) Apache Spark

97. Question: What is the term for a data quality issue in which data values are inconsistent across different sources?
a) Data duplication
b) Data inconsistency
c) Data aggregation
d) Data transformation
Answer: b) Data inconsistency

98. Question: Which data storage technology is designed for managing and querying data in a columnar format?
a) Apache Cassandra
b) Hadoop HDFS
c) Amazon Redshift
d) Apache HBase
Answer: c) Amazon Redshift

99. Question: What is the concept of "Data Stewardship" in the context of big data governance?
a) The practice of creating data visualizations for business insights
b) The responsibility for ensuring data accuracy and compliance with regulations
c) The process of transforming unstructured data into structured format
d) The practice of replicating data for high availability
Answer: b) The responsibility for ensuring data accuracy and compliance with regulations

100. Question: What is the main challenge associated with the "Velocity" aspect of big data?
a) Data volume is too large

mcqxpert.blogspot.com

b) Data variety is complex
c) Data is generated at a fast rate
d) Data is unstructured
Answer: c) Data is generated at a fast rate

EXPLORE MCQS ON OTHER TOPICS

mcqxpert.blogspot.com